

Stress assignment without morphology in Portuguese: the qualityless vowel /0/ and the Metrically Marked Item

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Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works (Makino 2010, 2012) on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We put forward a prosody-only account of Portuguese word stress. The analysis reduces Portuguese stress to the interaction of **a qualityless vowel /0/ bearing one mora + metrically marked items (MMIs) + general metrical theory with OT constraints**. This /0/ is the musical "ghost note" — **beat** (temporal position) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern): e.g. /0/ in *contador* /kont-a-dor-0/ vs. *contadora* /kont-a-dor-a/. MMIs are a class of **morpheme-sized elements** that, **when the condition is satisfied, render the final syllable of the prosodic word extrametrical**.

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when an MMI satisfies the condition)

Metrification is computed strictly from phonological information, together with a single piece of non-categorical lexical information — namely, the **MMI specification**. No reference to syntactic categories or to morphological bracketing is made at any point in the stress assignment process. This approach therefore **eliminates morphology from stress computation proper** and unifies items such as *contador*, *contadora*, *falava*, *falávamos*, *faláveis*, *número*, *numeroso*, *numérico*, *sobre* and *após* within a single minimal mechanism.

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0. Abbreviations, symbols and diacritics

PrWd = prosodic word; **Ft** = foot; **HdFt** = head foot; **MMI** = *metrically marked item*; **σ_{final_MMI}** = the syllable whose nucleus is the final vowel of an MMI; **L** = light syllable; **H** = heavy syllable

FTBIN = Parse a syllable into a binary foot; **TROCHEE** = Parse a syllable into a trochaic foot, where the first syllable is strong and the other(s) weak; **ALIGN-R(Ft, PrWd)** = align the right edge of each foot with the prosodic word; **ALIGN-R(HdFt, PrWd)** = align the right edge of the head foot with the prosodic word; **ALIGN-R(Ft, σ_{final_MMI})** = align the right edge of each foot with the syllable whose nucleus is the final vowel of an MMI; **ALIGN-R(HdFt, σ_{final_MMI})** = align the right edge of the head foot with the syllable whose nucleus is the final vowel of an MMI; **PARSE(σ)** = parse every syllable into a foot; ***CLASH** = avoid a stress clash; **#WSP** = Weight-to-Stress Principle at the left edge of the prosodic word

/ / = underlying representation; [] = surface representation; → = derivation; - = morpheme boundary; # = word boundary; /0/ = qualityless vowel bearing one mora; **C** = consonant; **Ç** (e.g. ʀ, ɳ, ʂ) = syllabic consonant that arises from the association of a consonant with the mora contributed by /0/; **μ_o** = the mora supplied by /0/; . = syllable boundary; ' = primary stress; , = secondary stress; **σ** = syllable; () = footing

1. Introduction

While previous accounts have relied on morphological information in stress computation, the present approach **dispenses with morphology and derives stress solely from phonological information** — syllable sequence, syllable number, syllable weight (at the left edge) and MMI.

2. The qualityless vowel /0/

A **qualityless** vowel /0/ that bears **one mora** in underlying representations is assumed — e.g. *contador* /kont-a-dor-0/ (cf. *contadora* /kont-a-dor-a/); *jacaré* /zakare-0/; *trabalhar* /trabaʎ-a-r0/; *trabalharmos* /trabaʎ-a-r0-mos/; *após* /a-pɔ0s/; *até* /ate0/. This vowel /0/ is the musical "**ghost note**" — **beat** (temporal position) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern). It is a purely phonological entity, serving various morphological purposes: in *contador* /kont-a-dor-0/ and *jacaré* /zakare-0/, the /0/ functions as the thematic vowel — stem vowel or *índice temático* (stem marker) — just as /a/ does in *contadora* /kont-a-dor-a/. On the other hand, it forms part of the infinitive or future subjunctive suffix /-r0/ in *trabalhar* /trabaʎ-a-r0/, and likewise occurs in the preposition *até* /ate0/, which does not display any internal

morphological structure.

The postulation of /0/ is not an ad hoc assumption but a necessary requirement of a stress theory based on the metrical foot. Strong–weak alternation requires at least two distinct temporal positions. Without such positions, alternation in strength cannot occur. Conversely, distinct temporal positions do not presuppose strength. It follows that temporal positions can exist independently of strength. In other words, **distinct temporal positions are a prerequisite for strong–weak alternation**, but not vice versa. If a stress pattern is defined, metrical-theoretically, as the *iterative alternation of strong and weak beats along the time axis*, then both strong and weak beats require distinct temporal positions. If the main stress — the most prominent alternation of strong and weak beats — falls on *-dor* in *contador*, **there must necessarily exist within *-dor* a temporal position corresponding to the weak beat paired with the strong beat**. In this way, positing /0/ is grounded in both **the primacy of temporal positions for strong–weak alternation and the very definition of the stress pattern**.

If /0/ is not assumed, *-dor* would constitute a degenerate foot — which, within metrical theory, is by definition a monosyllable left over at the end of footing — and this imperfect foot would serve as the head of the prosodic word, bearing the highest prominence. (This outcome would be theoretically undesirable, since footing in Portuguese is assumed to proceed from the right edge, as shown in Sections 4–6, and yet its initial foot would paradoxically be degenerate — a state of affairs for which metrical theory may offer no principled explanation.)

3. The metrical marked item: MMI

It is also assumed that there exists a class of *metrically marked items* (MMIs). These are **morpheme-sized forms** that are lexically specified to **render the final syllable of the PrWd extrametrical** (i.e. excluded from footing), but **only when their own final vowel serves as the nucleus of that penultimate syllable** — e.g. *-ic- /-ik-/* in *académico* /akadēm-**ik**-o/; *-vel /-vel/* in *agradável* /a-grad-a-**vel**-0/; *numer- /numér-/* in *número* /numér-o/; *-va- /-va-/* in *trabalhávamos* /trabaʎ-a-**va**-mos/; *-r- /-r0-/* in *trabalhardes* /trabaʎ-a-**r0**-des/. **Whether a morpheme belongs to this class is independent of its morphosyntactic category.**

4. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

possibilidade /pɔs-i-bil-i-dad-e/ → pɔ.si.bi.li.da.de → (,pɔ.si)(,bi.li)(ˈda.de) → [,pu.siβi.liˈða:.ði]
contador /kont-a-dor-0/ → kon.ta.do.r̩ → (,kon.ta)(ˈdo.r̩) → [,kõ.teˈðo:r̩]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (,kon.ta)(ˈdo.ra) → [,kõ.teˈðo:.rɐ]
coração /korason-0/ → ko.ra.so.ɳ → (,ko.ra)(ˈso.ɳ) → [,ku.rɐˈsẽũ]
parabéns /para-ben-0-s/ → pa.ra.be.ɳs → (,pa.ra)(ˈbe.ɳs) → [,pɐ.rɐˈβẽĩ]
jacaré /zakare-0/ → ʒa.ka.rɛ.ɛ → (,ʒa.ka)(ˈrɛ.ɛ) → [,ʒɐ.kɐˈrɛ:]
trabalhamos /trabaɫ-a-mos/ → tra.ba.ɫa.mos → (,tra.ba)(ˈɫa.mos) → [,trɐ.βɐˈɫɐ:.muʃ]
trabalhais /trabaɫ-a-des/ → tra.ba.ɫa.des → (,tra.ba)(ˈɫa.des) → [,trɐ.βɐˈɫaiʃ]
trabalhar /trabaɫ-a-r0/ → tra.ba.ɫa.r̩ → (,tra.ba)(ˈɫa.r̩) → [,trɐ.βɐˈɫa:r̩]
trabalhei /trabaɫ-a-i/ → tra.ba.ɫa.i → (,tra.ba)(ˈɫa.i) → [,trɐ.βɐˈɫɐi]
trabalhou /trabaɫ-a-u/ → tra.ba.ɫa.u → (,tra.ba)(ˈɫa.u) → [,trɐ.βɐˈɫo:]
sobre /sobre/ → so.bre → (ˈso.bre) → [ˈso:.βri]

This suggests the following constraints in interaction: **FTBIN**; **TROCHEE**; **ALIGN-R(Ft, PrWd)**; **ALIGN-R(HdFt, PrWd)**. (Note that the syllabification of /0/ is hypothetical, and that various possibilities are theoretically conceivable. What is crucial from a metrical-theoretical perspective is that **the syllable containing the mora supplied by /0/ — henceforth μ_0 — is a separate syllable from those preceding and following it.**)

5. Marked metrification

If the **final vowel of an MMI** serves as the **nucleus of the penultimate syllable of the PrWd**, the same iterative formation of trochaic binary feet proceeds **from that penultimate syllable — rather than from the final syllable** — with the rightmost foot serving as the head foot and bearing primary stress; the final syllable is thus excluded from metrification and remains unstressed:

académico /akadem-**ik**-o/ → a.ka.dɛ.mi.ko → (,a.ka)(ˈdɛ.mi)**ko** → [,ɐ.kɐˈðɛ:.mi.ku]
agradável /a-grad-a-**vel**-0/ → a.gra.da.**ve.ɫ** → (,a.gra)(ˈda.**ve.ɫ**) → [,ɐ.ɣrɐˈða:.vɛɫ]
número /numɛr-o/ → **nu.mɛ**.ro → (ˈ**nu.mɛ**)ro → [ˈnu:.mi.ru]
túnel /tunel-0/ → **tu.ne.ɫ** → (ˈ**tu.ne.ɫ**) → [ˈtu:.nɛɫ]
sótão /sɔtan-o/ → **sɔ.tan**.o → (ˈ**sɔ.tan**)o → [ˈsɔ:.tẽũ]
trabalhá^{vamos} /trabaɫ-a-**va**-mos/ → tra.ba.ɫa.**va**.mos → (,tra.ba)(ˈɫa.**va**)mos
→ [,trɐ.βɐˈɫa:.vɐ.muʃ]
trabalhá^{veis} /trabaɫ-a-**va**-des/ → tra.ba.ɫa.**va**.des → (,tra.ba)(ˈɫa.**va**)des → [,trɐ.βɐˈɫa:.vɐiʃ]
trabalhardes /trabaɫ-a-**r0**-des/ → tra.ba.ɫa.**r̩**.des → (,tra.ba)(ˈɫa.**r̩**)des → [,trɐ.βɐˈɫar̩.ðiʃ]

By contrast, if the **final vowel of an MMI** does **not function as the nucleus of the penultimate syllable of the PrWd**, **unmarked metrification**, as described in Section 4, emerges:

atonicidade /a-tõn-**ik**-i-dad-e/ → a.tõ.ni.si.da.de → (,a.tõ)(,ni.si)(ˈda.de) → [,ɐ.tu.ni.siˈða:.ðĩ]
numeroso /numɐr-oz-o/ → nu.mɐ.ro.zo → (,nu.mɐ)(ˈro.zo) → [,nu.mĩˈro:.zu]
trabalhava /trabaʎ-a-**va**/ → tra.ba.ʎa.**va** → (,tra.ba)(ˈʎa.**va**) → [,trɐ.βɐˈʎa:.vɐ]
trabalhar /trabaʎ-a-**rõ**/ → tra.ba.ʎa.r → (,tra.ba)(ˈʎa.r) → [,trɐ.βɐˈʎa:r]

When a PrWd contains more than one MMI, metrification proceeds from the final syllable of the MMI whose final vowel function as the nucleus of the penultimate syllable of the PrWd:

numérico /numɐr-**ik**-o/ → nu.mɐ.ri.ko → nu(ˈmɐ.ri)ko → [nuˈmɐ:.ri.ku]

All of the above suggests the ranking: **ALIGN-R(Ft, σ_{final_MMI})** >> **ALIGN-R(Ft, PrWd)** and **ALIGN-R(HdFt, σ_{final_MMI})** >> **ALIGN-R(HdFt, PrWd)**. Note that whereas syllabification of segments refers not only to phonological information but also to morphological information, the metrification of syllable strings refers only to phonological information.

6. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification is assumed to proceed as follows. These generalisations are **provisional** and call for further investigation.

6.1 Five or more syllables: only the leftmost foot is ternary

Only the **leftmost** foot is **ternary**; **subsequent** feet remain **binary** trochees:

papelaria /papel-ari-a/ → pa.pe.la.ri.a → (,pa.pe.la)(ˈri.a) → [,pɐ.pĩ.lɐˈða:.ðĩ]
computador /konput-a-dor-õ/ → kon.pu.ta.do.r → (,kon.pu.ta)(ˈdo.r) → [,kõ.pu.tɐˈðo:r]
entendimento /entend-i-ment-o/ → en.ten.di.men.to → (,en.ten.di)(ˈmen.to) → [,ẽ.tẽ.diˈẽ:.tu]

The constraint observations and violations of the winner and the losers are as follows:

Winner #($\sigma\sigma$)...#: NO *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation

Loser #(σ)($\sigma\sigma$)...#: NO *CLASH violation; ONE PARSE(σ) violation; NO FTBIN violation

$\therefore$ PARSE(σ) $\gg$ FTBIN

Winner #($\sigma\sigma\sigma$)...#: NO *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation

Loser #(σ)($\sigma\sigma$)...#: ONE *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation

$\therefore$ *CLASH decides (tie-breaking).

This suggests the following constraints: PARSE(σ) $\gg$ FTBIN; *CLASH

6.2 Initial L-H exception: initial light syllable unfooted

If a **light initial syllable** is immediately followed by a heavy syllable, it remains **unfooted**:

levantamento /levant-a-ment-o/ $\rightarrow$ le.van.ta.men.to $\rightarrow$ le(, van.ta)('men.to) $\rightarrow$ [lĩ, vẽ.ta 'mẽ:.tu]

Note that optional unary footing of the initial light syllable is not excluded, as in *levantamento* /levant-a-ment-o/ $\rightarrow$ le.van.ta.men.to $\rightarrow$ (, le)(, van.ta)('men.to) $\rightarrow$ [lĩ, vẽ.ta 'mẽ:.tu].

The constraint observations and violations are as follows:

Winner # L(, H. σ)...#:

NO *CLASH violation; ONE PARSE(σ) violation; NO FTBIN violation; NO #WSP violation

Loser # (, L.H. σ)...#:

NO *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation; ONE #WSP violation

$\therefore$ #WSP $\gg$ PARSE(σ) — since PARSE(σ) $\gg$ FTBIN, as inferred in Section 6.1, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

Winner #L(, H. σ)...#:

NO *CLASH violation; ONE PARSE(σ) violation; NO FTBIN violation; NO #WSP violation

Loser #(σ)(, L.H. σ)...#:

ONE *CLASH violation; NO PARSE(σ) violation; ONE FTBIN violation; NO #WSP violation

$\therefore$ *CLASH $\gg$ PARSE(σ) — since PARSE(σ) $\gg$ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This suggests the following constraint ranking: *CLASH, #WSP $\gg$ PARSE(σ) $\gg$ FTBIN

6.3 Three syllables: initial syllable unfooted

The **initial** syllable is **not footed**:

peessoa /peso-a/ → **pe**.so.a → **pe**('so.a) → [**pi**'so:.ɐ]
doutores /dout-or-0-s/ → **dou**.to.ɾs → **dou**('to.ɾs) → [**do**'to:.ɾiɸ]
pequeno /peken-o/ → **pe**.ke.no → **pe**('ke.no) → [**pi**'ke:.nu]
levar /lɛv-a-r0/ → **lɛ**.va.ɾ → **lɛ**('va.ɾ) → [**li**'va:ɾ]
até /atɛ0/ → **a**.tɛ.ɛ → **a**('tɛ.ɛ) → [**ɐ**'tɛ:]
após /apɔ0s/ → **a**.pɔ.ɔs → **a**('pɔ.ɔs) → [**ɐ**'pɔ:f]

Note that optional unary footing of the initial syllable, as in *levar* /lɛv-a-r0/ → lɛ.va.ɾ → (lɛ)('va.ɾ) → [li'va:ɾ], is not excluded. The constraint observations and violations of the winner candidate and the loser candidates are as follows:

Winner #σ('σσ)#: NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation

Loser #(.σ)('σσ)#: **ONE *CLASH violation**; NO PARSE(σ) violation; ONE FTBIN violation

∴ *CLASH >> PARSE(σ) — since PARSE(σ) >> FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This also suggests the following constraint ranking: *CLASH, #WSP >> PARSE(σ) >> FTBIN

7. The constraint set (provisional)

From the discussion in Sections 4–6, the following set of constraints and their rankings are inferred:

*CLASH, #WSP >> PARSE(σ) >> FTBIN;
TROCHEE;
ALIGN-R(Ft, σ_{final}_MMI) >> ALIGN-R(Ft, PrWd);
ALIGN-R(HdFt, σ_{final}_MMI) >> ALIGN-R(HdFt, PrWd)

8. Plausibility of the postulation of /0/

8.1 Historical corroboration: from Latin to Portuguese

Comparing the Portuguese nouns *amador* /am-a-dor-0/ and *razão* /razon-0/ with their Latin etymons *amatore* /amā-t-ōr-e-m/ and *rationem* /rat-iōn-e-m/ shows that the Portuguese /0/

corresponds to the Latin segment /e/. Many other examples of this kind can also be found. This suggests that, in this case, **/0/ was established as the result of the Latin /e/ having its quality deleted while its quantity was preserved.**

8.2 Synchronic corroboration: Plural in *-es* [-i] as the */-0s/* realisation

We analyse plural nouns and adjectives ending in *-es*, such as *contadores*, as /kont-a-dor-**0**-s/, with the thematic vowel */-0/*. In the singular *contador* /kont-a-dor-**0**/, μ_0 is assumed to be associated with the final rhotic */-r/* during syllabification, yielding [ˌkõ.tɐ'ðo:r] at the surface level. In the plural, on the other hand, the */0/* in the sequence */-C0s/#* surfaces ultimately as [i] through the linking of μ_0 to this inserted vocalic quality (orthographic *-es*), yielding the surface form [ˌkõ.tɐ'ðo:ri]. This analysis avoids positing an underlying /e/ in the plural — as in /kont-a-dor-e-s/ — which would have to be deleted in the singular /kont-a-dor-e/ at the surface level. Such a move would be incompatible with the many lexical items that retain final */-e/* [-i] — e.g. *árvore*, *are*, *hectare*, *Zêzere*.

8.3 Northern EP corroboration: paragoge

In Northern varieties, words ending in */-r0/* or */-l0/* may surface as [-ri] ~ [-ri] or [-li] ~ [-li]. For instance, the infinitive *comer* /kõm-e-r**0**/ may surface as [ku'me:ri] or [ku'me:ri], with paragogic [i] or [i], rather than as the standard [ku'me:r]. In the standard variety, μ_0 is associated with [r], whereas in those Northern varieties μ_0 is associated with the epenthetic quality [i] or [i], with **the stress pattern remaining unchanged**. In other words, the "ghost note" */0/* thus occasionally acquires inserted timbre, becoming a "full note" [i] or [i], without nevertheless altering the rhythm. This indicates that μ_0 , **irrespective of its association with any quality, consonantal or vocalic, constitutes the weak beat in a strong–weak alternation.**

8.4 Cross-Romance corroboration: Portuguese vs. Spanish

The main stress position of the future subjunctive forms *trabalharmos* and *traballar* can be determined solely on the basis of phonological information, by analysing the **future subjunctive suffix -r** as the **MMI */-r0/* with */0/***:

trabalharmos /trabaʎ-a-r**0**-mos/ → tra.ba.ʎa.ɾ.mos → (ˌtra.ba)(ˈʎa.ɾ)mos
traballar /trabaʎ-a-r**0**/ → tra.ba.ʎa.ɾ → (ˌtra.ba)(ˈʎa.ɾ)

This future subjunctive suffix is **-re /-re/ with /e/** in Spanish, which is, like **-r /-r0/** in Portuguese, an **MMI** and is assumed to render the final syllable of the PrWd extrametrical:

trabajáremos /trabax-a-re-mos/ → tra.ba.xa.re.mos → (,tra.ba.)('xa.re)mos
trabajare /trabax-a-re/ → tra.ba.xa.re → (,tra.ba.)('xa.re)

From this, it becomes clear that, **in stress computation, the Portuguese "ghost note" /0/ functions in exactly the same way as the Spanish "full note" /e/**. The stress patterns of the Portuguese noun *coração* and verbe *falar* and their Spanish cognates *corazón* and *hablar* can likewise be derived in parallel by positing /0/:

coração /korason-0/ → ko.ra.so.ŋ → (,ko.ra)('so.ŋ) → [,ku.rɐ'sẽũ]
corazón /koraθon-0/ → ko.ra.θo.ŋ → (,ko.ra)('θo.ŋ) → [,ko.ra'θɔn]
falar /fal-a-r0/ → fa.la.ɾ → fa('la.ɾ) → [fɐ'laːɾ]
hablar /habl-a-r0/ → a.bla.ɾ → a('bla.ɾ) → [a'βlar]

The difference between these two languages lies primarily in the degree of complexity of the phonological processes that apply after stress assignment, from the intermediate representations (,ko.ra)('so.ŋ) and (,ko.ra)('θo.ŋ), fa('la.ɾ) and a('bla.ɾ), respectively. Thus, **the model proposed in this paper may be considered extendable to Spanish as well.**

8.5 Theoretical parsimony: economy of representation

Under the theoretical framework of this paper, the stress-pattern opposition between the quasi-minimal pair *também* and *lambem* can be easily explained by the final /0/ in *também* /tan-ben-0/, which is absent from *lambem* /lanb-e-n/:

também /tan-ben0/ → tan.be.ŋ → tan('be.ŋ) → [tẽ'bẽĩ]
lambem /lanb-e-n/ → lan.ben → ('lan.ben) → ['lẽː.bẽĩ]

The stress-pattern difference can thus be attributed solely to the difference in their segmental sequences. Without positing /0/, by contrast, one would have to assign additionally a lexically specified accent to *também*, thereby requiring two types of information in the lexicon: segmental and lexical-accentual.

9. Orthographic corollary

When **marked metrification** emerges due to an MMI, the stressed nucleus is **typically indicated orthographically** by an acute accent or a circumflex accent, since the outcome **deviates from the default stress position** — e.g. *âmbar*, *côdea*, *número*, *pêssego*, *sótão*, *túnel*, *académico*, *agradável*, *trabalhávamos*, *trabalháveis*. This close correlation suggests that **speakers' prosodic intuitions** about **stress location** are reflected in the standard orthography, thereby providing independent support for the analysis.

10. Conclusion

Thus, Portuguese stress can be fully accounted for by **the interaction of:**

- **The qualityless vowel /0/ bearing one mora**
- **Metrically marked items (MMIs)**
- **General metrical theory formalised through OT constraints**

without recourse to morphologically-driven stress assignment. The principles and parameters of primary stress assignment in Portuguese are as follows:

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when an MMI satisfies the condition)

In this way, Portuguese stress patterns are reduced to two possibilities in terms of primary stress placement, yielding a relatively simple system. However, these patterns are superimposed on a wide variety of morphological structures, and it is the attempt to describe stress assignment from the perspective of those morphological structures that appears to have contributed to the observed complexity.

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